

UNIT-3

MAC Sub Layer

MAC Sublayer → The MAC sublayer (Media Access Control sublayer) is a part of the Data Link Layer (Layer 2) in networking, especially in the OSI Model.

Where it fits.

The Data Link Layer is divided into two sublayers.

- LLC (Logical Link Control)
- MAC (Media Access Control).

What the MAC Sublayer does

→ The MAC sublayer is responsible for controlling how devices on a network gain to the transmission medium (like Ethernet cables or WiFi).

Functions

- ★ Hardware addressing → Uses MAC addresses (unique physical addresses assigned to network interfaces).

★ Frame Formatting

- Encapsulates data into frames for transmission.
- Adds headers and trailers (including source and destination MAC addresses).

★ Media Access Control

- Decides when a device can transmit data.
- Prevents collisions in shared media networks.

Examples → Ethernet uses CSMA/CD
→ Wifi uses CSMA/CA

★ Error Detection

uses mechanisms like CRC (Cyclic Redundancy Check)

MAC Addressing

MAC addressing means using a MAC address to identify devices on a local network.

A MAC address is a unique physical address assigned to a device's network interface card (NIC), such as Ethernet or wifi adapter.

It is 48-bit hardware address used in the Data link Layer (Layer 2) of the OSI model.

Format of MAC Address

A MAC address is written in hexadecimal form.

Example

- 00:1A:2B:3C:4D:5E

- AA-BB-CC-DD-EE-FF

It contains 6 bytes (48 bits)

Structure of MAC Address.

A MAC address has two parts:-

1. First 24 bits - Manufacturer ID (OUI - Organizationally Unique Identifier)
 - Identifies the company that made the device.
2. Last 24 bits - Device ID
 - Unique number assigned by manufacturer.

Functions of MAC Addressing.

1. Device Identification Each device on a LAN can be uniquely identified.
2. Local Communication - used to send data between devices in the same network.
3. Frame Delivery - Switches use MAC addresses to forward frames to correct destinations.

Binary Exponential Backoff

Binary Exponential Backoff is a method used in computer networks to reduce collisions when multiple devices try to send data at the same time.

It is commonly used in Ethernet networks with CSMA/CD (Carrier Sense Multiple Access with Collision Detection).

Why it is Needed

When two or more devices transmit simultaneously, a collision occurs. After collision, devices must wait before sending again.

If they retry immediately, another collision may happen.

Binary Exponential Backoff (BEB) helps by choosing a random waiting time.

How it works.

After each collision,

1. Device stops transmission.
2. It waits for a random time.
3. Then it tries to transmit again.

Formula

After n collisions, device chooses random number K :

$$0 \leq K \leq 2^n - 1$$

Then waiting time -
 $K \times \text{slot time}$

Example $\rightarrow$ First collision ($n=1$)

$$K = 0 \text{ or } 1$$

wait for 0 or 1 slot.

Second collision ($n=2$)

$$K = 0 \text{ to } 3$$

wait for 0, 1, 2 or 3 slots.

Third collision ($n=3$)

$$K = 0 \text{ to } 7$$

wait increases more

Binary $\rightarrow$ Uses powers of 2

Exponential $\rightarrow$ Range doubles after each collision.

Advantages.

- Reduces repeated collisions.
- Improves network efficiency.
- Fair chance for all devices.

Disadvantages
→ After many collisions, waiting time becomes longer.

Distributed Random Access Schemes / Contention Schemes

These are network access methods where many devices share the same communication channel.

- Each device tries to send data independently.
- Because many devices use the same channel, collision may occur.

These schemes are called

- Distributed - no central controller
- Random Access - any station can transmit anytime
- Contention Scheme - stations compete for channel access.

Types of Random Access / Contention Schemes.

ALOHA

RGIPV 2025

① Pure ALOHA.

② Slotted ALOHA.

ALOHA

ALOHA protocol is a random access protocol used in computer networks for transmitting data through a shared communication channel.

In ALOHA, any station can send data whenever it wants without asking permission from other stations.

- Because many stations share the same channel, two or more stations may transmit at the same time, causing a collision.

Need of ALOHA.

When multiple devices use the same communication channel:-

- there must be a method to control access
- devices should know when to send data.
- collisions must be handled properly.

ALOHA provides a simple solution to this problem.

Types of ALOHA.

1. Pure ALOHA
2. Slotted ALOHA

Pure ALOHA Imp

Pure ALOHA is a random access protocol used in computer network.

It is a basic version of ALOHA protocol. In which a station transmits data whenever it has data to send without checking whether the channel is busy or free.

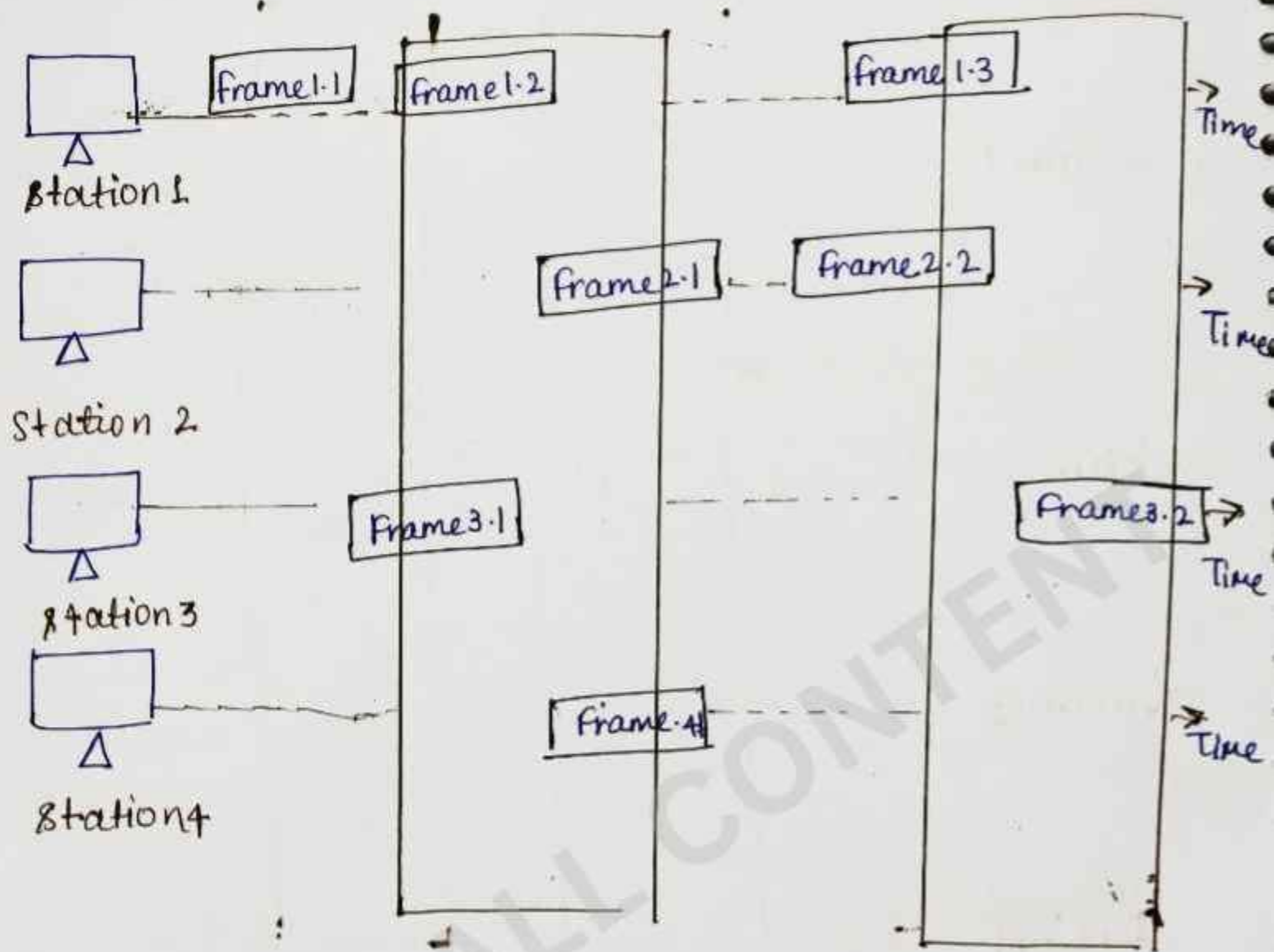
Pure ALOHA is a protocol ~~where~~ ^{working}.

A node sends a frame whenever it has data.

The receiver sends an acknowledgment (ACK) if successful.

If no ACK is received (collision assumed), the sender :-

- Waits for a random back off time.
- Retries transmissions



Frames in Pure ALOHA.

Pure ALOHA uses the channel efficiency only about 18.4% of the time.

A better version is Slotted ALOHA.

Slotted ALOHA Imp

Slotted ALOHA is an improved version of Pure ALOHA used in computer network to reduce collision.

In slotted ALOHA.

- Time is divided into equal slots.
- Each slot = time to send one frame
- A device can transmit only at the beginning of a slot.

★ This simple rule reduces collision a lot.

Working -

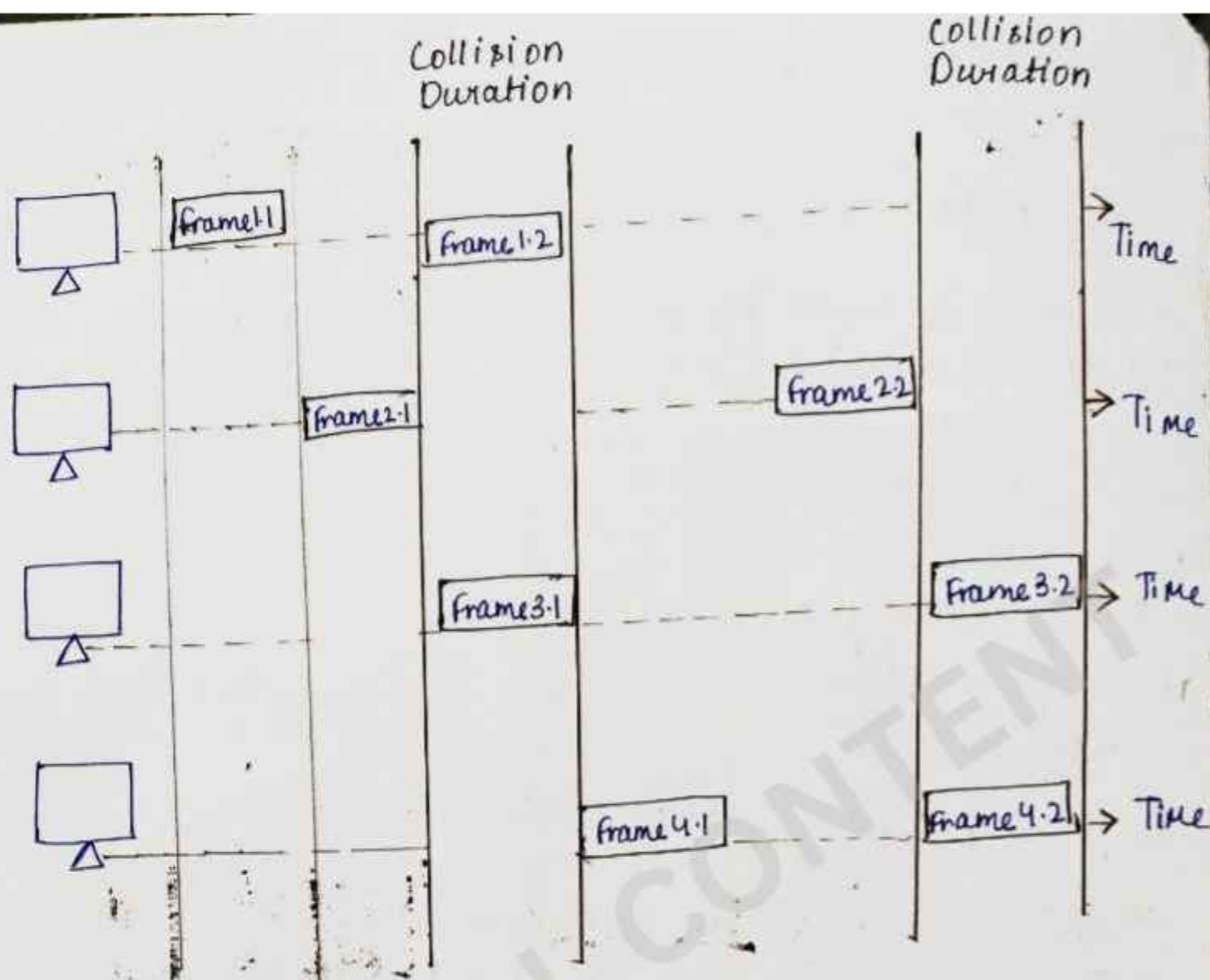
When a station has data, it waits for the next time slot.

It sends the frame at start of the slot.
If:-

- Only one station sends $\rightarrow$ success
- More than one sends $\rightarrow$ collision.

★ After collision:-

- waits for a random number of slots
- Try again.



Frames in Slotted ALOHA

This is double the efficiency of Pure ALOHA
 $(18.4\%) = 36.8\%$.

Simple Example

- Pure ALOHA $\rightarrow$ Student speaks anytime $\rightarrow$ chaos.
- Slotted ALOHA $\rightarrow$ Students speak only when teacher says 'start' $\rightarrow$ more organized.

CSMA, CSMA/CD, CSMA/CA

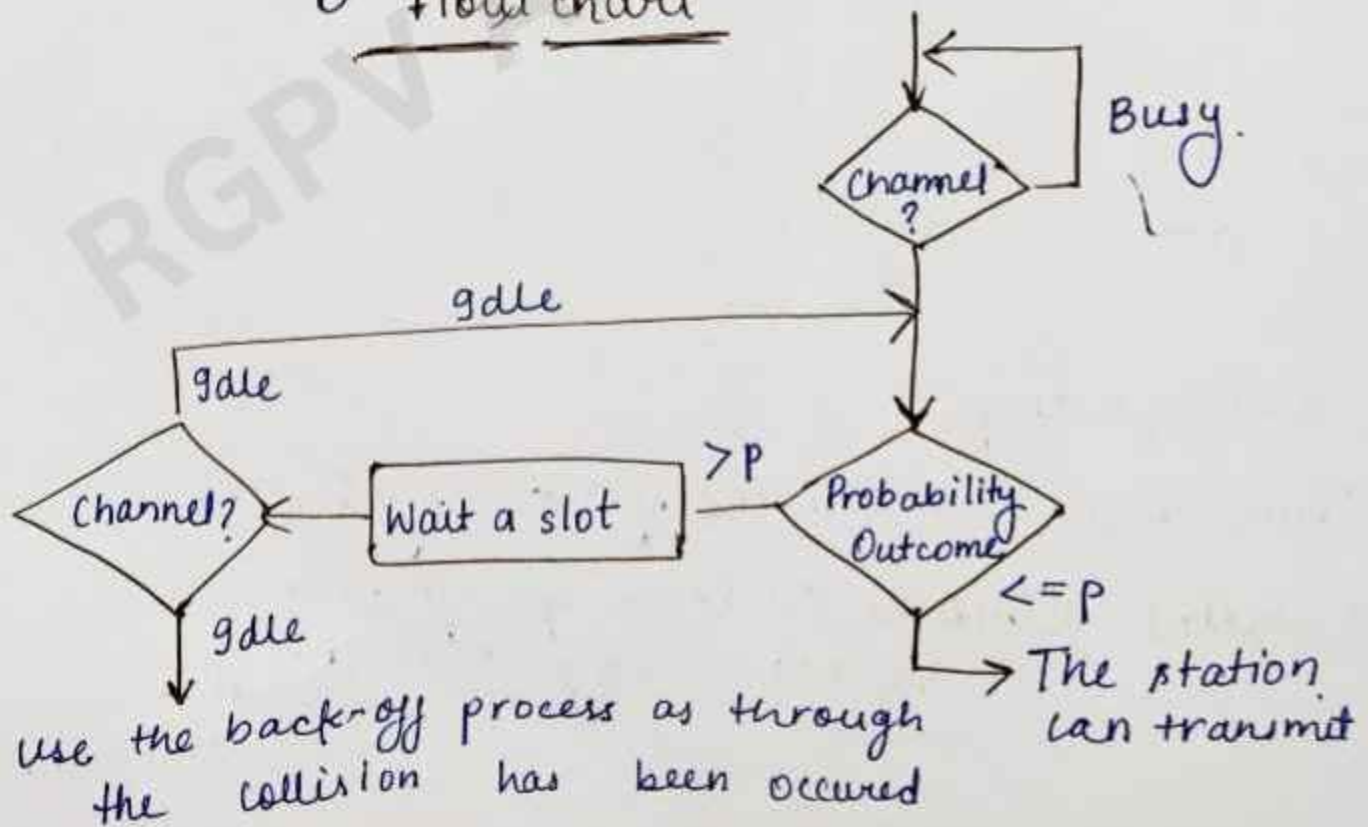
CSMA (Carrier Sense Multiple Access)

CSMA is a medium access control protocol in which a station senses the channel before transmitting data to reduce the chances of collision.

Basic concept:

- Multiple devices share a common communication channel.
- Before sending data, each device:
 - listens (sense) to the channel.
 - Sends only if the channel is idle (free)

Flowchart



Types of CSMA

① 1-Persistent CSMA

In this method, a device continuously checks the channel.

Working

1. If channel is free - send data immediately (probability = 1)
2. If channel is busy $\rightarrow$ keep waiting and checking continuously.
3. As soon as channel becomes free $\rightarrow$ transmit instantly.

Advantage.

Simple and fast.

Disadvantage

High chance of collision because many devices may transmit together.

② Non-Persistent CSMA.

In this method the device does not continuously check the channel.

Working.

1. If channel is free $\rightarrow$ transmit data.
2. If channel is busy $\rightarrow$ wait for a random time.
3. After ~~waiting~~ waiting, check again.

Advantages

- fewer collisions

Disadvantages

- more delay because of random waiting

③

p-Persistent CSMA

used mainly in slotted channels.

when channels becomes free -

- Transmit with probability p
- wait for next time slot with probability $1-p$

Example if $p = 0.6$.

- 60% chance to transmit
- 40% chance to wait.

Advantages

- Balance collision and delay

Disadvantages

- More complex than other methods

CSMA/CD 2025

[Carrier Sense Multiple Access with Collision Detection]

CSMA/CD :-

- CSMA/CD is a media access control method used in traditional Ethernet LANs.
- It helps multiple computers share the same communication channel without major data conflicts.
- When many devices are connected to the same cable, two devices may try to send data at the same time.
- This causes a collision. CSMA/CD detects and handles these collision.

working -

- ① Before sending data, the device checks whether the channel is busy or free.
 - if channel is free $\rightarrow$ transmit data
 - if channel is busy $\rightarrow$ wait.
- ② The device starts transmitting data when the channel becomes free.

③ If another device also transmits at the same time:

- signals overlap
- Collision occurs.

• The device immediately detects the collision.

④ After detecting collision:-

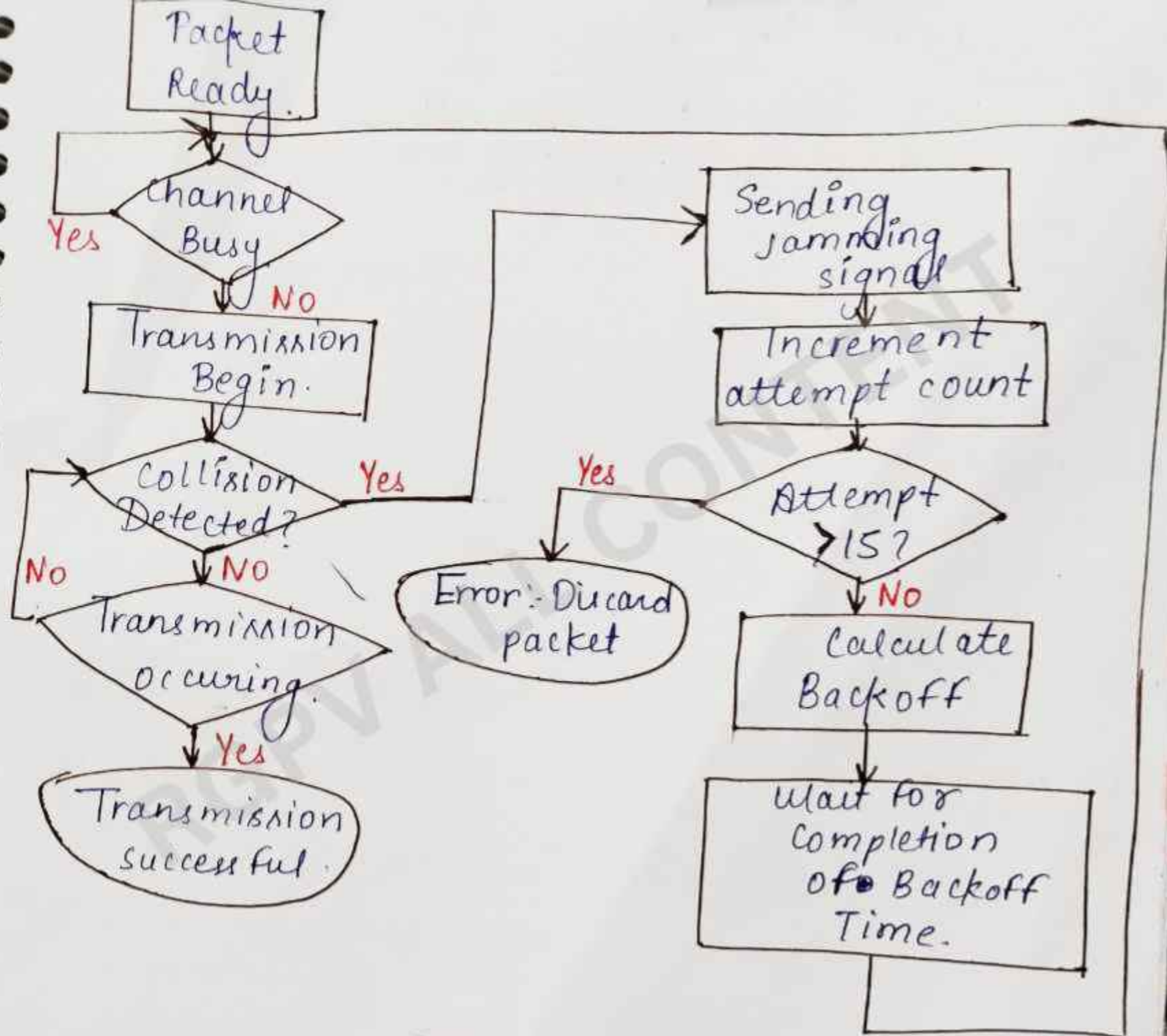
- Device stop transmitting.
- A jam signal is sent to inform all devices that collision occurred.

⑤ Each devices waits for a random amount of time before retransmitting

- This random waiting reduces the chance of another collision.

⑥ After waiting:

- Device again checks the channel
- If free → retransmit data.



CSMA/CA Imp

(Carrier Sense Multiple Access with Collision Avoidance).

CSMA/CA is a network communication protocol used in wireless networks to control how multiple devices share the same communication channel without causing collisions.

It mainly used in Wifi networks based on IEEE 802.11 standards.

Working

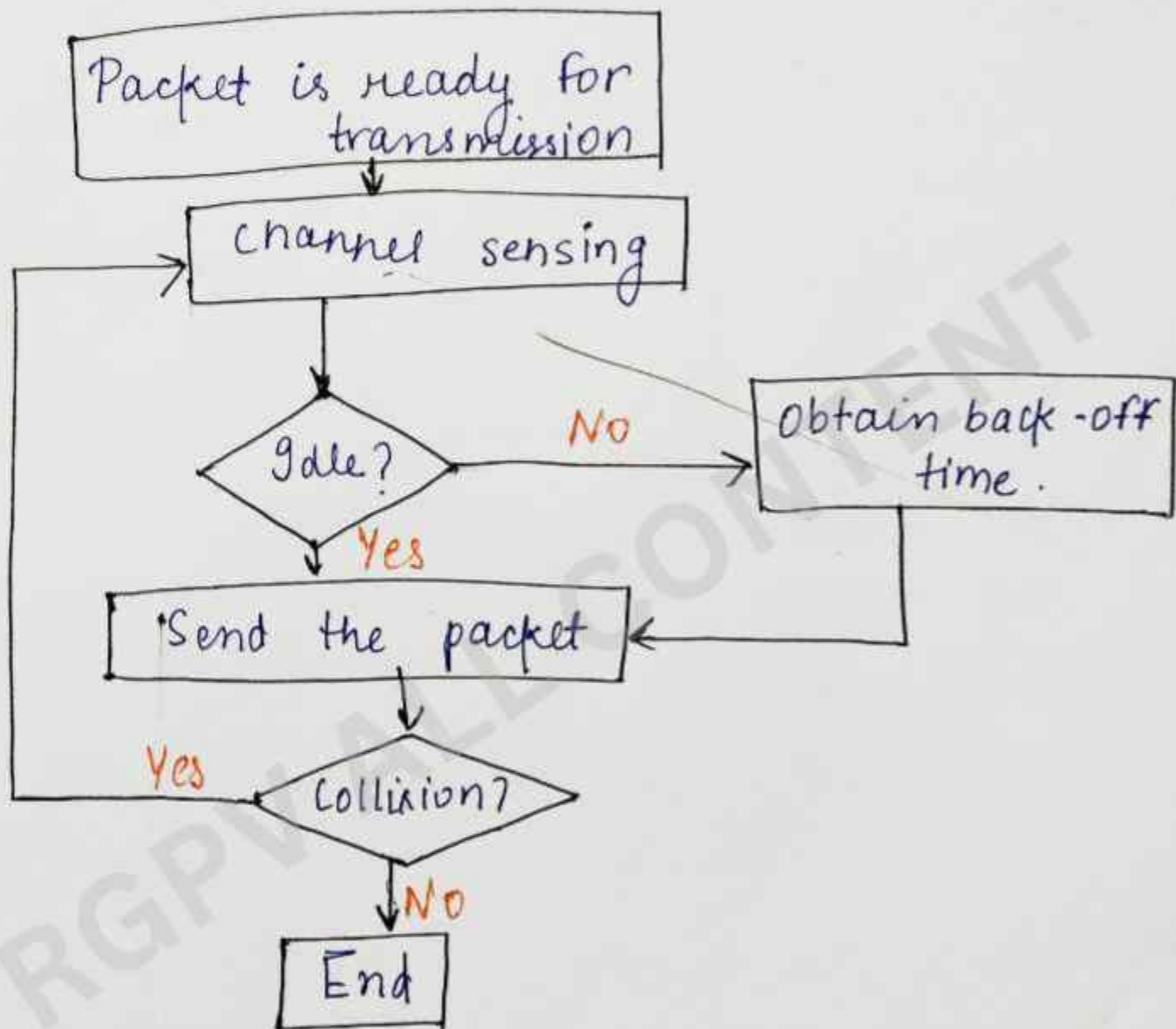
In this method, before sending data, a device first checks whether the communication channel is free or busy.

If the channel is busy, the device wait.

If the channel become free, the device still waits for a small random time called the back off time before transmitting data.

This process helps in avoiding collisions between data packets.

Collision Avoidance → because it tries to prevent collisions before they happen.



CSMA/CA

Collision free Protocols

Collision free protocols are network protocols to make sure that data collisions do not happen during transmission.

Types.

- ① Basic Bit map.
- ② BRAP.
- ③ Binary Count Down.

① Basic Bit Map

It is designed for a shared communication channel where many computers (stations) use the same medium to send data.

The main purpose of this protocol is - To avoid collision while transmitting data.

Simple Idea - Suppose there are N stations (computers) connected to one channel.

Each station gets.

- A fixed number
- like 0, 1, 2, 3, ...

Before sending data, every station tells whether it wants to transmit or not using a bitmap.

How it works

Bitmap Transmission.

Each station sends 1 bit in order:

- 1 $\rightarrow$ "I want to send data"
- 0 $\rightarrow$ "I do not want to send data"

This sequence of bit is called the bitmap.

Example

Suppose there are 4 stations.

station	want to send data?	Bit
0	Yes	1
1	No	0
2	Yes	1
3	No	0

No collision occurs because everyone already knows whose turn it is:-

Advantages.

1. Completely collision free
2. Fair access to all stations
3. Efficient under heavy load.

Disadvantages.

1. Slow when many stations exist
2. Wastes time if few stations want to send.

BRAP

BRAP stands for Bit-Map Reservation Access Protocol.

It is a collision-free protocol used in channel allocation.

In this each station uses a reservation bit to indicate whether it wants to transmit data. and stations then send data in an orderly manner without collisions

Working of BRAP

- Suppose there are N stations connected to a network
- Each station gets 1 bit in a reservation map

- If a station wants to send data, it puts its bit as 1.
- If it does not want to send data, it puts 0.
- After all bits are sent.
 - The stations that marked 1 send their data one by one in order.
 - Since transmission order is fixed, collision does not occur.

Example

Suppose 4 stations are present

Station	Want to send?	Bit
S1	Yes	1
S2	No	0
S3	Yes	1
S4	No	0

Reservation bits = 1010

So,

- S1 sends first
- S3 sends second

★ No collision happens.

Advantages

- Collision free.
- Fair access to channel.
- Efficient under heavy load.

Disadvantages

- Reservation bits create overhead.
- Slow when few stations want to send data.

③ Binary Count Down

Binary Countdown protocol is a collision-free protocol used to control access to a shared communication channel.

It allows multiple stations to compete for the channel without causing collision.

In this stations compete for channel access by transmitting their binary address bit-by-bit.

The station with the highest priority (highest binary address) wins and gets permission to send data without collision.

Looping -

Every station has a unique binary address. Stations send address bits starting from the most significant bit (MSB).

If a station sends 0 but detects 1 on the channel, it stops competing.

The station with the highest binary number remains active and transmits data.

Example

Addresses:-

A = 1010

B = 1001

C = 1110

Since 1110 is highest, station C wins the channel.

Important Points

- Uses binary priorities
- Higher binary number = Higher priority
- No data collision occurs

Advantages

- 1-Collision free communication
- 2-Fast channel allocation
- 3-Efficient under heavy traffic
- 4-Simple priority - based selection.

Disadvantages.

- 1 Stations with low priority may wait for a long time
- 2 High priority ~~channel~~ stations can dominate the channel.
- 3 Requires stations to know binary priorities

Limited Contention Protocol. 2023

Limited Contention Protocol is a protocol used in the MAC (Medium Access Control) sublayer to reduce collisions in a shared communication channel.

It contains the advantages of:

- Contention Protocols $\rightarrow$ simple and fast at low traffic.
- Collision Free Protocols - efficient at high traffic.

Basic Idea

$\rightarrow$ In normal contention methods like CSMA, many stations try to send data at the same time causing collisions.

In collision-free methods, collisions are avoided but the process becomes slower.

So, Limited Contention Protocol limits the number of stations competing at one time.

This :-
reduce collision
improves efficiency
and gives better performance under heavy load.

Adaptive Tree Walk Protocol.

- Adaptive Tree walk protocol is a type of Limited Contention Protocol used in computer networks to reduce collisions in a shared communication channel.
- It divides stations into groups in the form of a tree structure.
- If collision occurs, the group is divided into smaller ~~go~~ subgroups until only one station remains and successfully transmits data.

Basic Idea

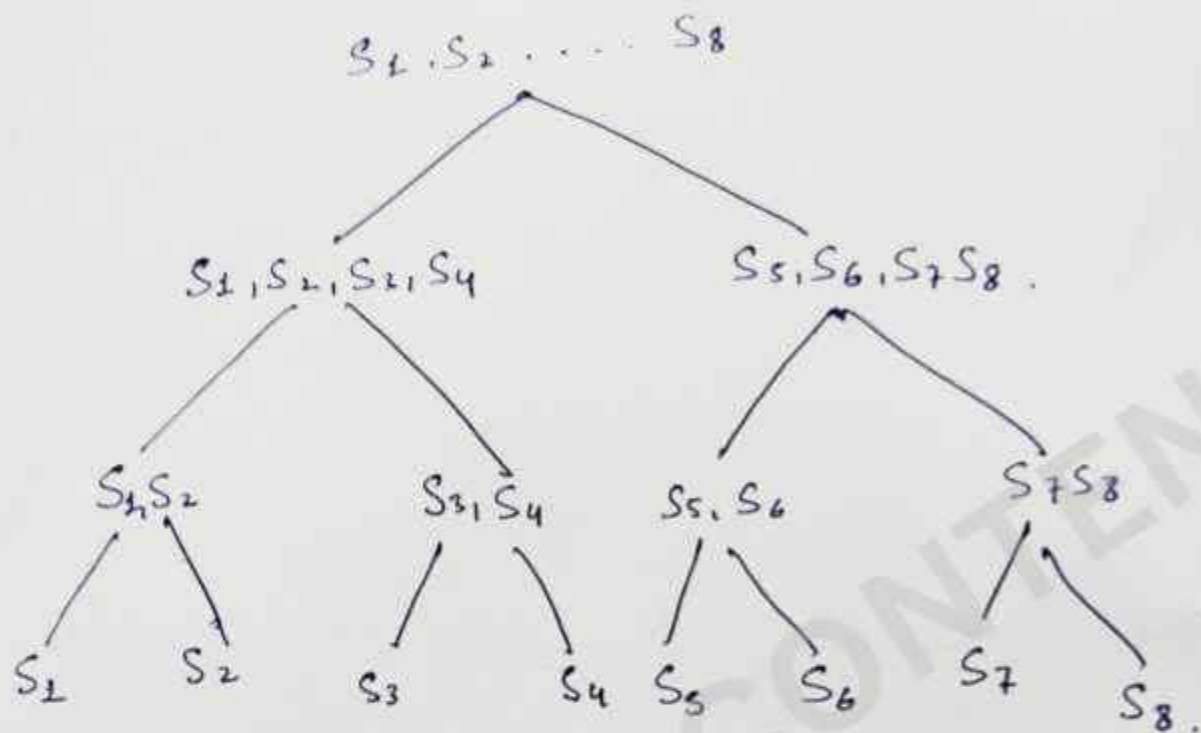
Instead of allowing stations to compete together:

- Stations are divided into groups.
 - Only one group is allowed to transmit.
 - If collision occurs - divide the group further.
 - Continue until successful transmission happens.
- This reduces collision significantly.

ATWP would organize the stations using the following tree.

Suppose we have 8 stations.

$[S_1, S_2, \dots, S_8]$



Why it is called 'Adaptive'.

It is called Adaptive because.

- the grouping changes dynamically according to collisions.
- and the protocol adapts to network traffic conditions.

Applications

Used in :

- LAN Networks.
- Shared communication channels.
- High traffic Network System

Advantages

1. Reduces collisions
2. Efficient during heavy traffic
3. Better channel utilization
4. Fair transmission opportunity

★ Adaptive tree walk work like a binary tree

Disadvantages.

1. More complex
2. Extra delay due to subgroup formation
3. Requires tree management

Performance Measuring Metrics.

Performance Measuring Metrics are the parameters used to check how well a networking is working.

They help us measure the speed, quality, efficiency, and reliability of communication.

1. Bandwidth

Bandwidth means the maximum amount of data that can be transferred through a network in a given time.

- Measured in : bps (bits per second)
- Higher bandwidth $\Rightarrow$ more data transfer.

Example $\Rightarrow$ A 100 Mbps connection can transfer more data than a 10 Mbps connection.

$$\text{Bandwidth} = \frac{\text{Amount of data}}{\text{Time}}.$$

2. Throughput

Throughput means the actual amount of data successfully transferred over the network.

- Real performance of network
- Always less than or equal to bandwidth.

Example $\Rightarrow$ Bandwidth = 100 Mbps
Actual data received = 80 Mbps.
Then Throughput = 80 Mbps.

3 Latency (Delay)

Latency is time taken by data to travel from sender to receiver.

- Measured in milliseconds (ms).
- Lower latency is better

Example $\Rightarrow$ Online gaming require less latency.

4 Jitter Jitter means the variations in Packet delay.

If packets do not arrive at equal time intervals. jitter occurs.

Example.

During video calls, voice breaking happens because of high jitter.

5. Packet loss Packets loss occurs when some data packets are lost during transmission.

Causes

- Network congestion
- Hardware failure
- Weak signal.

Effect

- Slow internet
- Video buffering
- Call Interruption

IEEE Standards 802 series & their variant 2023

IEEE 802 standards are a group of networking standards developed by the IEEE for LAN and MAN communication.

These standards define how devices communicate in wired and wireless networks.

Main IEEE 802 Standards.

① • IEEE 802.2 (Logical Link Layer - LLC)

- IEEE 802.2 is a networking standard developed by IEEE.
- It defines the logical link control (LLC) sublayer of the Data Link Layer in OSI model.
- IEEE 802.2 works at the DLL of OSI model. It is placed above the MAC layer and below the Network layer.
- The main purpose of IEEE 802.2 is to provide reliable communication between devices & it helps in managing data transfer and controlling communication errors.

- The LLC layer checks for transmission errors during communication.
- It helps in detecting corrupted or lost data frames.

2] IEEE 802.3 [Ethernet]

- IEEE 802.3 is a networking standard developed by IEEE.
- It defines Ethernet technology used for wired Local Area Network (LANs).
- The main purpose of IEEE 802.3 is to provide communication between devices in a wired network.
- It defines rules for data transmission through cable.
- IEEE 802.3 uses wired transmission media for communication.
- It supports twisted pair cable, coaxial cable, and fibre optic cable.
- It uses a method called CSMA/CD (Carrier Sense Multiple Access with Collision Detection). This method helps devices share the same communication channel efficiently.

3) IEEE 802.5 (Token Ring)

- It defines the Token Ring Network technology used in LAN.
- The main purpose of IEEE 802.5 is to provide organized and collision-free communication between devices.
- It controls how devices access and share the network medium.
- IEEE 802.5 is based on Token Ring technology in this network. devices are connected in a logical ring structure.
- It uses a special data packet called a token. only the device holding the token is allowed to send data on the network.
- Since only one device can transmit data at a time, collisions do not occur. This makes communication more organized and efficient.

4) IEEE 802.11

It defines the rules for wireless communication in LAN.

The main purpose of IEEE 802.11 is to provide wireless internet and network communication. It allows devices to connect without using physical cables.

IEEE 802.11 is commonly known as Wi-Fi Technology.

It is widely used in homes, schools, offices, airports, and public places.

IEEE 802.11 uses radio waves for communication between devices. It enables laptop, smartphones, and tablets to connect wirelessly.

It mainly uses 2.4 GHz and 5 GHz frequency bands. and newer version may also use the 6 GHz band for faster communication.